

Project-Based Learning: Applied Projects for Student Engagement

Presented on Friday: 10/11/2024
@



PRESENTED BY
 **HAWKES**
LEARNING



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“The **Joy** of a **Teacher** is the **Success** of his **Students**.” – Samuel Dominic Chukwuemeka

Applied Projects should be: I-S-M-A-R-T

I: Interdisciplinary

Some form of writing is required.

The writing should be part of the assessment.

S: Specific

What is the specific topic?

What are the specific concepts?

What are the specific objectives?

M: Measurable

How do we measure the success of the project?

How is the project evaluated?

A: Authentic

Also known as Active Learning, the project must be a real-world project that connects the knowledge acquired in the classroom to applications outside the classroom.

R: Realistic

The project should not be too simple, neither should it be too complex. At least one of the objectives of the project should be from the course learning objectives.

T: Timely

The project should have firm deadlines. It is okay to have more than one due date. However, clear expectations of deliverables for each due date should be communicated. If it is a group project or a project that needs to be completed in phases, then firm deadlines are required for each phase of the project.

Arithmetic: Number Theory: ISBN

One of the applications of Number Theory is in Modular Arithmetic.

One of the applications of Modular Arithmetic is in the calculation of the Check Digit of an **International Standard Book Number (ISBN)**.

ISBN stands for **International Standard Book Number**.

It is a **unique identifier** for any **published book**.

The **Check Digit** of an **International Standard Book Number (ISBN)** is the **last digit** of that number.

For ISBN-10, the check digit is the 10th digit.

For ISBN-13, the check digit is the 13th digit.

Objectives

Students will:

- (1.) Perform basic arithmetic operations (addition, subtraction, multiplication, and division) on integers.
- (2.) Solve applied problems on numbers.
- (3.) Discuss the concepts of: Divisibility, Factors, and Multiples.
- (4.) Meet one of the learning objectives of the VCCS (Virginia Community College System) standards for:

[MTH 111: Basic Technical Mathematics](#)

*(Provides a foundation in mathematics with emphasis in **arithmetic**, unit conversion, basic algebra, geometry and trigonometry.).*

Objectives (Continued)

MTH 133: Mathematics for Health Professions

*(Presents in context the **arithmetic** of fractions and decimals, the metric system and dimensional analysis, percents, ratio and proportion, linear equations, topics in statistics, topics in geometry, logarithms, topics in health professions including dosages, dilutions and IV flow rates.).*

MTH 154: Quantitative Reasoning

[Solve real-life problems requiring interpretation and comparison of various representations of ratios (i.e., fractions, decimals, rates, and percentages including part to part and part to whole, per capita data, growth and decay via absolute and relative change)].

(5.) Meet the QM (Quality Matters) and USDOE (United States Department of Education) requirements for distance education as regards the provision of RSI (Regular and Substantive Interaction).

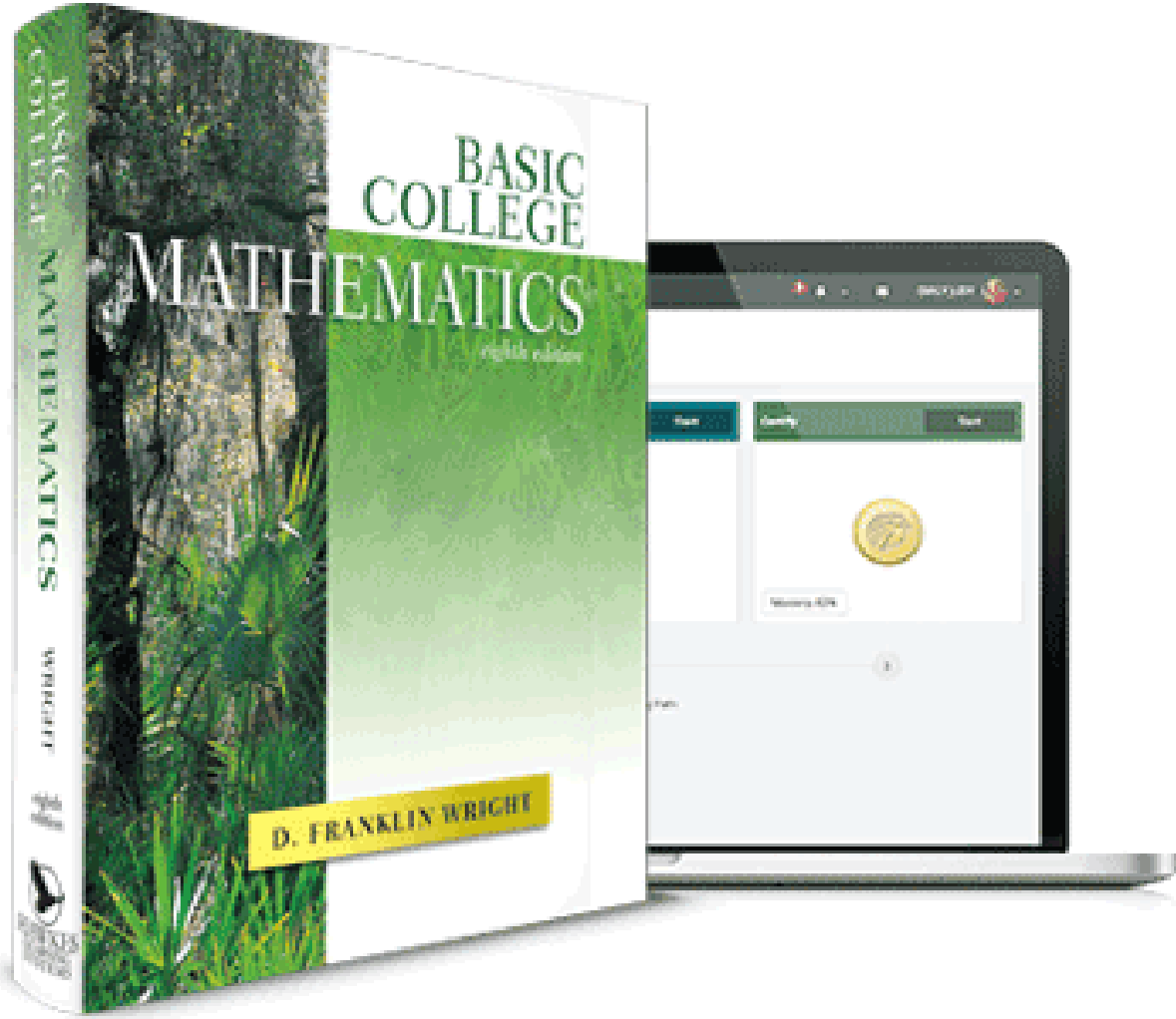
Federal Register: Distance Education and Innovation

St. John's University: New Federal Requirements for Distance Education: Regular and Substantive Interaction (RSI)

Student – Content Interaction: Very high

Student – Student Interaction: Flexible

Student – Faculty Interaction: Flexible



Basic College Mathematics

by D. Franklin Wright

Basic Mathematics helps students develop basic arithmetic skills, as well as reasoning and problem-solving skills. The material emphasizes the why and how of basic operations and procedures to give students greater meaning to their learning.

Formats: Software, Textbook, eBook

Product	ISBN
Software + eBook	978-1-944894-11-5
Software + eBook + Textbook	978-1-944894-13-9

Specific Objectives

- (1.) Verify the check digit for ISBN-10
- (2.) Verify the check digit for ISBN-13
- (3.) Convert ISBN-10 to ISBN-13
- (4.) Convert ISBN-13 to ISBN-10

Objectives

The **Show Me Your Receipts** project is designed to:

(1.) Meet one of the learning objectives of the VCCS (Virginia Community College System) standards for:

MTH 132: Business Mathematics

*(Provides instruction, review, and drill in **percentage**, cash and trade discounts, mark-up, payroll, sales, property and other taxes, simple and compound interest, bank discounts, loans, investments, and annuities).*

MTH 133: Mathematics for Health Professions

*(Presents in context the arithmetic of fractions and decimals, the metric system and dimensional analysis, **percents**, ratio and proportion, linear equations, topics in statistics, topics in geometry, logarithms, topics in health professions including dosages, dilutions and IV flow rates.).*

MTH 154: Quantitative Reasoning

*[Solve real-life problems requiring interpretation and comparison of various representations of ratios (i.e., fractions, decimals, rates, and **percentages** including part to part and part to whole, per capita data, growth and decay via absolute and relative change)].*

Objectives (Continued)

(2.) Meet the QM (Quality Matters) and USDOE (United States Department of Education) requirements for distance education as regards the provision of RSI (Regular and Substantive Interaction).

[Federal Register: Distance Education and Innovation](#)

[St. John's University: New Federal Requirements for Distance Education: Regular and Substantive Interaction \(RSI\)](#)

Student – Content Interaction: Very high

Student – Student Interaction: Flexible

Student – Faculty Interaction: Flexible

(3.) Solve applied problems involving percents.

Case 0: What Item?

SamDom4Peace did not pay a sales tax in one of the items.

Identify the item.



WAL*MART
HARRISONBURG, VA
ST# 1726 OP# 9036 TE# 36 TR# 295

SM ORNGE BA	060504947825 F	4.98
ENH SERIES	004138800435	5.47

	SUBTOTAL	10.45
TAX		0.00
TAX		0.05
	TOTAL	10.50
VISA DEBIT TEND		10.50
VISA	**** * 5412	
	CHANGE DUE	0.00

ITEMS SOLD 2

TC# 6656 9149 3374 8396 4532



03/05/24 17:39:25

Case 1:

Given: Initial price and Tax rate

Calculate:

- (a.) Tax
- (b.) Total price

Walmart

Save money. Live better.

ANNISTON, AL

ST# 329 OP# 9045 TE# 45 TR# 6459

TC# 8921 2351 6518 0520 3650

EL 6PK REPE 008597200414 23.96

SUBTOTAL23.96

TAX 1 10 %

TOTAL

VISA DEBIT TEND

VISA **** * 6892

CHANGE DUE0.00

ITEMS SOLD 1

03/08/23 10:27:41

Case 2:

Given: Tax rate and Total price

Calculate:

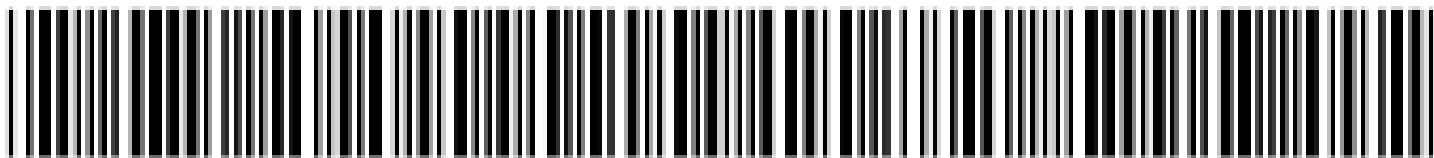
(a.) Initial price (Subtotal)

(b.) Tax



HARRISONBURG, VA
ST# 2807 OP# 9032 TE# 32 TR# 6785

TC# 1220 4925 8325 2083 4409



BODY	001111101251	████████
LOTION	007989617481	████████
RG AERO TWI	001700006835	████████
SC M MEM	009650647307	████████

SUBTOTAL		████████
TAX 1	5.3 %	████████
TOTAL		34.64
VISA DEBIT TEND		34.64
VISA *****		3802
CASH BACK		20.00
CHANGE DUE		20.00

ITEMS SOLD 4

02/02/23 18:11:28

(b.) Total price



(d.) Unit price of the fifth item per pound



Case 5:
Given: Tax, Sam's Club credit, Remaining balance (Use \$11.1798)

- Calculate:
- (a.) Total price
 - (b.) Initial price (Subtotal)
 - (c.) Tax rate



Self Checkout
(434) 978 - 2122
CHARLOTTESVILLE, VA
03/06/24 16:44 8979 06365 093 9093

SAMUEL

E	990325124	MMCOHOFILLEF		R
		SUBTOTAL		
		TAX		0.16
		TOTAL		
		SAM'S CASH TEND		4.96
		DEBIT TEND		11.18
		CHANGE DUE		0.00

EFT DEBIT PAY FROM PRIMARY
11.18 TOTAL PURCHASE

Case 6: As at March 2024, the City of Harrisonburg in the State of Virginia charges 1% sales tax on raw foods and fruits and a high tax rate on cooked food.

(I guess they want Virginians to cook their meals.)

Chukwuemeka was hungry and decided to buy hot food at Walmart.

The hot food is high in calories and is a source of carbohydrates (carbs).

He also decided to buy a bag of fresh oranges to provide essential vitamins.

The receipt shows three tax rates: the tax rate on the oranges (shown) and two different tax rates on the hot food (not shown).

Given:

(I.) Initial price of the bag of oranges

(II.) Initial price of the hot food

(III.) Tax rate on the bag of oranges

(IV.) Tax on the bag of oranges

(V.) 1st tax on the hot food (Use \$0.39591)

(VI.) 2nd tax on the hot food (Use \$0.5229)

Calculate:

(a.) Subtotal (Sum of the initial prices)

(b.) 1st tax rate on the hot food

(c.) 2nd tax rate on the hot food

(d.) Total price

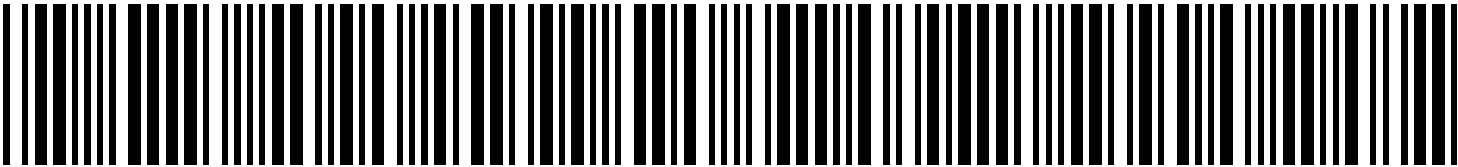


WAL*MART
HARRISONBURG, VA
ST# 1726 OP# 9036 TE# 36 TR# 289

SM ORNGE BA	060504947825	F	4.98
HOT FOOD	026285200000		7.47
SUBTOTAL			12.45
TAX			0.40
TAX	2	1 %	0.05
TAX			0.52
TOTAL			13.42
VISA DEBIT TEND			
VISA	****	****	5412
CHANGE DUE			0.00

ITEMS SOLD 2

TC# 1656 9049 3374 8998 4572



03/05/24 17:30:43

Case 7:

Samuel placed two different online orders on two different days on the Amazon website.

The two different orders were sent to the same address in the State of Virginia.

None of the orders has food or drinks.

However, the orders were charged different tax rates.

- (a.) Calculate the tax rate for the 1st order.
- (b.) Calculate the tax rate for the 2nd order.
- (c.) Samuel asked you to explain why the tax rates were different.
 - (i.) Ask Samuel at least one question in that regard.
 - (ii.) Provide at least one explanation to Samuel.

1 st Order	2 nd Order																												
Place your order By placing your order, you agree to Amazon's privacy notice and conditions of use. Order Summary <table><tr><td>Items (3):</td><td>\$47.92</td></tr><tr><td>Shipping & handling:</td><td>\$7.36</td></tr><tr><td>Free Shipping:</td><td>-\$7.36</td></tr><tr><td>Total before tax:</td><td>\$47.92</td></tr><tr><td>Estimated tax to be collected:*</td><td>\$2.38</td></tr><tr><td>Total:</td><td>\$50.30</td></tr><tr><td>Gift Card:</td><td>-\$5.00</td></tr><tr><td>Order total:</td><td>\$45.30</td></tr></table> Qualifying offers: Free Shipping How are shipping costs calculated?	Items (3):	\$47.92	Shipping & handling:	\$7.36	Free Shipping:	-\$7.36	Total before tax:	\$47.92	Estimated tax to be collected:*	\$2.38	Total:	\$50.30	Gift Card:	-\$5.00	Order total:	\$45.30	Place your order By placing your order, you agree to Amazon's privacy notice and conditions of use. Order Summary <table><tr><td>Items (2):</td><td>\$45.78</td></tr><tr><td>Shipping & handling:</td><td>\$7.90</td></tr><tr><td>Free Shipping:</td><td>-\$7.90</td></tr><tr><td>Total before tax:</td><td>\$45.78</td></tr><tr><td>Estimated tax to be collected:</td><td>\$2.43</td></tr><tr><td>Order total:</td><td>\$48.21</td></tr></table> Qualifying offers: Free Shipping How are shipping costs calculated?	Items (2):	\$45.78	Shipping & handling:	\$7.90	Free Shipping:	-\$7.90	Total before tax:	\$45.78	Estimated tax to be collected:	\$2.43	Order total:	\$48.21
Items (3):	\$47.92																												
Shipping & handling:	\$7.36																												
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Total:	\$50.30																												
Gift Card:	-\$5.00																												
Order total:	\$45.30																												
Items (2):	\$45.78																												
Shipping & handling:	\$7.90																												
Free Shipping:	-\$7.90																												
Total before tax:	\$45.78																												
Estimated tax to be collected:	\$2.43																												
Order total:	\$48.21																												

(b.) Initial price of the 2nd item



Case 9:

Given:

- (I.) Initial price
- (II.) Discount
- (III.) Percent Discount
- (IV.) Sale price

But:

The percent discount on the label is an approximate value.
Calculate the exact percent discount.



Objectives

The **Measurements and Units** project is designed to:

(1.) Meet one of the learning objectives of the VCCS (Virginia Community College System) standards for:

MTH 111: Basic Technical Mathematics

*(Provides a foundation in mathematics with emphasis in arithmetic, **unit conversion**, basic algebra, geometry and trigonometry.).*

MTH 130: Fundamentals of Reasoning

*(Presents elementary concepts of algebra, linear graphing, financial literacy, descriptive statistics, and **measurement** & geometry.).*

MTH 131: Technical Mathematics

*(Presents algebra through **unit conversion**, trigonometry, vectors, geometry, and complex numbers.).*

MTH 133: Mathematics for Health Professions

*(Presents in context the arithmetic of fractions and decimals, the **metric system and dimensional analysis**, percents, **ratio and proportion**, linear equations, topics in statistics, topics in geometry, logarithms, topics in health professions including **dosages, dilutions and IV flow rates**.).*

MTH 154: Quantitative Reasoning

[Solve real-life problems requiring interpretation and comparison of various representations of ratios (i.e., fractions, decimals, rates, and percentages including part to part and part to whole, per capita data, growth and decay via absolute and relative change)].

(2.) Meet the QM (Quality Matters) and USDOE (United States Department of Education) requirements for distance education as regards the provision of RSI (Regular and Substantive Interaction).

Federal Register: Distance Education and Innovation

St. John's University: New Federal Requirements for Distance Education: Regular and Substantive Interaction (RSI)

Student – Content Interaction: Very high

Student – Student Interaction: Flexible

Student – Faculty Interaction: Flexible

(3.) Convert between International Metric System of units and United States Customary System of units.

Container Used: Soap container

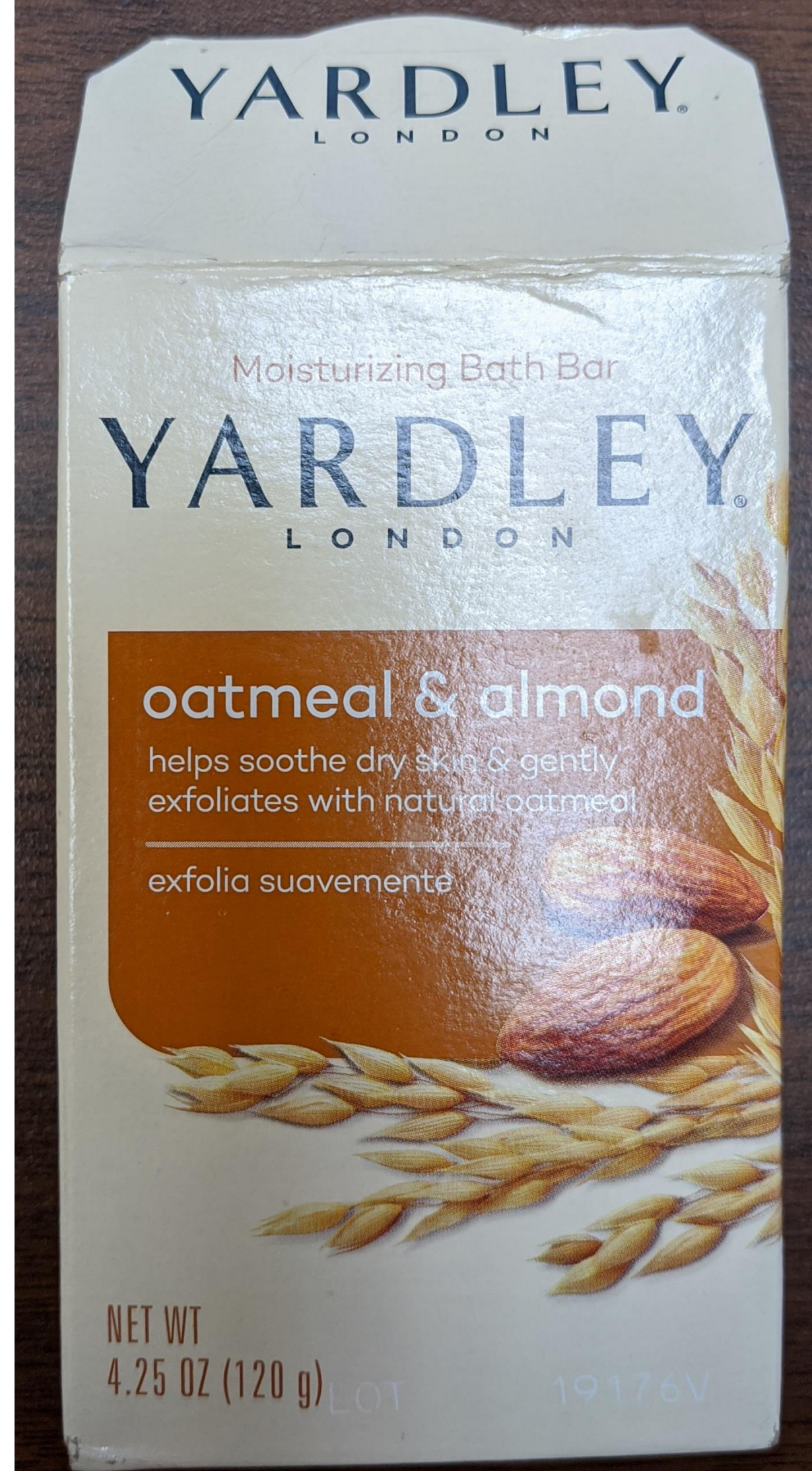
Calculations:

First Conversion: Convert 4.25 ounce to gram.

Second Conversion: Convert 120 grams to ounce.

Only Required Tables

Students Projects



[PreCalculus: Piecewise Function: Power Bill](#)

[PreCalculus: Piecewise Function: Water Bill](#)

[PreCalculus: Piecewise Function: Federal Tax](#)

Objectives

Students will:

- (1.) Discuss piecewise functions.
- (2.) Discuss real-world applications of piecewise-functions.
- (3.) Solve problems involving the application of piecewise functions.

Objectives (Continued)

(4.) Meet one of the learning objectives of the VCCS (Virginia Community College System) standards for:

[MTH 161: PreCalculus I](#)

(Presents topics in power, polynomial, rational, exponential, and logarithmic functions, and systems of equations and inequalities.).

[MTH 167: PreCalculus with Trigonometry](#)

(Presents topics in power, polynomial, rational, exponential, and logarithmic functions, systems of equations, trigonometry, and trigonometric applications, including Law of Sines and Cosines, and an introduction to conics.).

(5.) Meet the QM (Quality Matters) and USDOE (United States Department of Education) requirements for distance education as regards the provision of RSI (Regular and Substantive Interaction).

[Federal Register: Distance Education and Innovation](#)

[St. John's University: New Federal Requirements for Distance Education: Regular and Substantive Interaction \(RSI\)](#)

Student – Content Interaction: Very high

Student – Student Interaction: Flexible

Student – Faculty Interaction: High

Example: Alabama Power

Develop a piecewise function for the Billing Months: June – September.

Define variables: Let:

p = power consumption per KWh (KiloWatt hour)

c = cost of power consumption per KWh

$$c(p) = \begin{cases} 0.12233p + 14.5; & 0 \leq p \leq 1000 \\ 0.124859p + 11.971; & p > 1000 \end{cases}$$

[Students Projects](#)

Screenshot of Family Dwelling: Residential Service rates of Alabama Power.

Source: <https://www.alabamapower.com/content/dam/alabama-power/pdfs-docs/Rates/FD.pdf>

MONTHLY RATE	
Base Charge: \$14.50 per customer; plus	
Charge for Energy:	
BILLING MONTHS JUNE - SEPTEMBER 12.2330¢ per kWh for the first 1000 kWh, plus 12.4859¢ per kWh for all over 1000 kWh.	BILLING MONTHS OCTOBER - MAY 12.2330¢ per kWh for the first 750 kWh, plus 11.0330¢ per kWh for all over 750 kWh.
MINIMUM BILL	
No monthly bill shall be less than \$14.50 plus applicable provisions of Rate T.	

As of the 7th day of July, 2018; the residential water rates of Calhoun County Water Authority residents is:

First 3,000 Gallons:	\$17.35 Minimum Per Month
Next 2,000 Gallons:	\$5.24 per 1,000 Gallons
Next 5,000 Gallons:	\$3.96 per 1,000 Gallons
All over 10,000 Gallons:	\$3.10 per 1,000 Gallons

Develop a piecewise function for the residential water rates.

Define variables: Let:

g = gallons of water (in gallons)

r = rate (in dollars per gallons)

$$r(g) = \begin{cases} 17.35; & 0 \leq g \leq 3000 \\ 0.00524g + 1.63; & 3000 < g \leq 5000 \\ 0.00396g + 8.03; & 5000 < g \leq 10000 \\ 0.0031g + 16.63; & g > 10000 \end{cases}$$

Federal Income Tax 2018

The 2018 Federal Income Tax rates for the four filing options is shown in the table below.

Using the variable: p as the taxable income and t as the tax, develop a piecewise function for each filing option.

Tax Rate	Single	Head of household	Married Filing Jointly Or Qualifying Widow	Married Filing Separately
10%	Up to \$9,525	Up to \$13,600	Up to \$19,050	Up to \$9,525
12%	\$9,526 to \$38,700	\$13,601 to \$51,800	\$19,051 to \$77,400	\$9,526 to \$38,700
22%	\$38,701 to \$82,500	\$51,801 to \$82,500	\$77,401 to \$165,000	\$38,701 to \$82,500
24%	\$82,501 to \$157,500	\$82,501 to \$157,500	\$165,001 to \$315,000	\$82,501 to \$157,000
32%	\$157,501 to \$200,000	\$157,501 to \$200,000	\$315,001 to \$400,000	\$157,001 to \$200,000
35%	\$200,001 to \$500,000	\$200,001 to \$500,000	\$400,001 to \$600,000	\$200,001 to \$300,000
37%	\$500,001 or more	\$500,001 or more	\$600,001 or more	\$300,001 or more

Single

$$t(p) = \begin{cases} 0.1p; & 0 \leq p \leq 9525 \\ 0.12p - 190.5; & 9525 < p \leq 38700 \\ 0.22p - 4060.5; & 38700 < p \leq 82500 \\ 0.24p - 5710.5; & 82500 < p \leq 157500 \\ 0.32p - 18310.5; & 157500 < p \leq 200000 \\ 0.35p - 24310.5; & 200000 < p \leq 500000 \\ 0.37p - 34310.5; & p > 500000 \end{cases}$$

Head of Household

$$t(p) = \begin{cases} 0.1p; & 0 \leq p \leq 13600 \\ 0.12p - 272; & 13600 < p \leq 51800 \\ 0.22p - 5452; & 51800 < p \leq 82500 \\ 0.24p - 7102; & 82500 < p \leq 157500 \\ 0.32p - 19702; & 157500 < p \leq 200000 \\ 0.35p - 25702; & 200000 < p \leq 500000 \\ 0.37p - 35702; & p > 500000 \end{cases}$$

$$t(p) = \begin{cases} 0.1p; & 0 \leq p \leq 19050 \\ 0.12p - 381; & 19050 < p \leq 77400 \\ 0.22p - 8121; & 77400 < p \leq 165000 \\ 0.24p - 11421; & 165000 < p \leq 315000 \\ 0.32p - 36621; & 315000 < p \leq 400000 \\ 0.35p - 48621; & 400000 < p \leq 600000 \\ 0.37p - 60621; & p > 600000 \end{cases}$$

$$t(p) = \begin{cases} 0.1p; & 0 \leq p \leq 9525 \\ 0.12p - 190.5; & 9525 < p \leq 38700 \\ 0.22p - 4060.5; & 38700 < p \leq 82500 \\ 0.24p - 5710.5; & 82500 < p \leq 157000 \\ 0.32p - 18270.5; & 157000 < p \leq 200000 \\ 0.35p - 24270.5; & 200000 < p \leq 300000 \\ 0.37p - 30270.5; & p > 300000 \end{cases}$$

PreCalculus: Matrix Application: Cryptology

One of the applications of Matrices is in Cryptology.

One of the applications of Cryptology is the encoding (encrypting/enciphering) and decoding (decrypting/deciphering) of secret messages.

Objectives

Students will:

(1.) Encode a secret message.

(2.) Decode the secret message.

(3.) Meet one of the learning objectives of the VCCS (Virginia Community College System) standards for:

MTH 161: PreCalculus I

*(Presents topics in power, polynomial, rational, exponential, and logarithmic functions, and **systems of equations** and inequalities.)*

MTH 165: Finite Math

*(Presents topics in **systems of equations**, **matrices**, linear programming, mathematics of finance, counting theory, probability, and Markov Chains.)*

Objectives (Continued)

MTH 167: PreCalculus with Trigonometry

*(Presents topics in power, polynomial, rational, exponential, and logarithmic functions, **systems of equations**, trigonometry, and trigonometric applications, including Law of Sines and Cosines, and an introduction to conics.)*

MTH 266: Linear Algebra

*(Covers **matrices**, vector spaces, determinants, **solutions of systems of linear equations**, basis and dimension, eigenvalues, and eigenvectors.)*

(4.) Meet the QM (Quality Matters) and USDOE (United States Department of Education) requirements for distance education as regards the provision of RSI (Regular and Substantive Interaction).

Federal Register: Distance Education and Innovation

St. John's University: New Federal Requirements for Distance Education: Regular and Substantive Interaction (RSI)

Student – Content Interaction: Very high

Student – Student Interaction: Very high (if the project is a group project)

Student – Faculty Interaction: High

2 by 2 Matrix

Given: A scenario between Samuel and his Mother.

Mother: Samuel my son, what did you learn in church today?

Samuel: Mother, the priest said something very important.

He told us to always

124 48 47 16 55 22 80 29 20 8

His message is encoded with the encoding matrix:

5 2 3 1

If you were the mother, and your son wrote this to you, what would you do? ☺

3 by 3 Matrix

Please complete this sentence.

If you, the world will be a good place to live.

The fill-in message is the encoded message:

49 83 91 25 50 30 33 41 91 28 51 52 17 26 40 33 48 84

The message is encoded with the encoding matrix:

1 2 2 1 1 3 1 2 1

Cipher Key

<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>
1	2	3	4	5	6
<i>G</i>	<i>H</i>	<i>I</i>	<i>J</i>	<i>K</i>	<i>L</i>
7	8	9	10	11	12
<i>M</i>	<i>N</i>	<i>O</i>	<i>P</i>	<i>Q</i>	<i>R</i>
13	14	15	16	17	18
<i>S</i>	<i>T</i>	<i>U</i>	<i>V</i>	<i>W</i>	<i>X</i>
19	20	21	22	23	24
<i>Y</i>	<i>Z</i>		?	!	.
25	26	27, 0	28	29	30, 0

Objectives

The **Inferential Statistics** project is designed to:

(1.) Meet the requirements of the VCCS detailed course outline for MTH 245

- Technology Application

- Construct statistical tables, charts, and graphs using appropriate technology.
- Calculate descriptive and inferential statistics using an appropriate statistical software package.
- **Complete statistical project.** Students are required to complete some form of semester project in their course that is worth a significant portion of the student's grade. This could be either an individual or group effort, and could be completed in stages through the semester or as a single, stand-alone exercise. As a minimum, the project should require students to manipulate and draw statistical inferences from a large, realistic data set using a statistical software package.

(2.) Meet one of the learning objectives of the VCCS (Virginia Community College System) standards for:

(a.) [MTH 155: Statistical Reasoning](#)

*(Presents elementary statistical methods and concepts including visual data presentation, descriptive statistics, probability, **estimation**, **hypothesis testing**, correlation and linear regression.).*

(b.) [MTH 245: Statistics I](#)

*(Presents an overview of statistics, including descriptive statistics, elementary probability, probability distributions, **estimation**, **hypothesis testing**, correlation, and linear regression.).*

Objectives (Continued)

(3.) Draw statistical inferences from a sample onto a population. This is achieved by any of these tasks:

- (a.) Estimate a population parameter using a sample statistic.
- (b.) Conduct a hypothesis test of a population parameter using a sample statistic.

(4.) Meet the QM (Quality Matters) and USDOE (United States Department of Education) requirements for distance education as regards the provision of RSI (Regular and Substantive Interaction).

[Federal Register: Distance Education and Innovation](#)

[St. John's University: New Federal Requirements for Distance Education: Regular and Substantive Interaction \(RSI\)](#)

Student – Content Interaction: Very high

Student – Student Interaction: Flexible

Student – Faculty Interaction: Very High

References

- Chukwuemeka, S.D (2024, January 30). *Samuel Chukwuemeka Tutorials: Math, Science, and Technology*. Retrieved from <https://conferencepresentations.appspot.com/>
- Alabama Power. (n.d.). Alabama Power. Retrieved from <https://www.alabamapower.com/content/dam/alabama-power/pdfs-docs/Rates/FD.pdf>
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- Calhoun County Water Association. (n.d.). Calhoun County Water Association. Retrieved from <http://www.calhouncwa.com/rates.htm>
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